

Du Duong Nguyen

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SKILLS

- Built and tested **C++ software components**, utilizing **data structures** and debugging techniques to improve program performance
- Designed responsive user interfaces using **JavaScript and HTML/CSS**, enhancing usability and user experience
- Collaborated with a team using **Git** for version control, managing branches, resolving merge conflicts, and coordinating code integration to support efficient and organized group development
- Applied strong **problem-solving skills** to debug and resolve technical issues, improving overall application stability

PROJECTS

Portfolio Website - [Github](#) - [Website](#)

May 2026 - Aug 2026

- Developed a responsive personal portfolio website using HTML and CSS to showcase projects, technical skills, and experience.
- Designed a clean, user-friendly interface with responsive layouts optimized for desktop and mobile devices.
- Implemented CSS animations and interactive UI elements to create a polished and engaging user experience.
- Structured reusable HTML and CSS components to maintain organized, scalable, and maintainable code.
- Applied modern web development practices, including semantic HTML, Flexbox, CSS Grid, and responsive design techniques.

Rubik's Cube - [Github](#)

Sep 2025 - Dec 2025

- Developed and implemented efficient cube-solving algorithms (e.g., BFS/heuristics) to compute optimal or near-optimal move sequences.
- Modelled Rubik's Cube state transitions using appropriate data structures and handled complex state space navigation.
- Implemented robust input validation and custom exception handling (e.g., `IncorrectFormatException`) to ensure resilience against malformed cube inputs.
- Used systematic debugging and unit testing (or test-driven approaches) to validate solver correctness across various cube configurations.

Address Book - [Github](#)

Jan 2025 - Apr 2025

- Implemented manual heap allocation (`malloc/realloc/free`) for a resizable array of Contact structs and their string fields, including error checking and safe cleanup.
- Built robust input routines that validate numeric fields (phone number, age), handle buffer/newline cleanup (`fgets`, `strcspn`, input flushing), and allow retry logic with limited attempts.
- Designed custom serialization/deserialization of contact records to/from text files, including both machine-readable (load/save) and human-readable report formats, with error handling for missing/invalid files.

Poco Loco game - [Github](#)

Sep 2024 - Dec 2024

- Built and maintained a Python script with clear logic, control flow, and idiomatic use of language features.
- Read, parsed, and wrote structured input/output (e.g., text/CSV) while handling edge cases and malformed data.
- Design step-by-step processing logic (loops, conditionals, and data transformations) to reliably solve the domain problem.
- Used debugging techniques and iterative testing to verify correctness across different scenarios and inputs.

WORKPLACE EXPERIENCE

• **Math Tutoring - Burnaby North Secondary School Volunteer**
Mar 2024

Burnaby, BC, Feb 2024 -

• **Retail - Shopper Drugs Mart Volunteer**
Nov 2023

Burnaby, BC, Oct 2023 -

• **Cashier and Closing - Queen Elizabeth Hospital Volunteer**
Aug 2022

Charlottetown, PEI, Jun 2021 -

EDUCATION

Simon Fraser University

Bachelor of Applied Science, Major in Computer Science

Burnaby, BC

Sep 2024 - Present